Fundamentals of Computer Graphics

Lecture 1

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Outline

- Introduction
- Major Areas in Computer Graphics(CG)
- Historical Milestones
- Application Areas in CG
- Computer Graphics Vs Computer vision
- Raster and Vector Graphics
- Graphics pipeline

Introduction to Computer Graphics

- What is Computer Graphics?
  - computer graphics describes any use of computers to create and manipulate images.
- The major product of computer graphics is a picture.
- Graphics can be two- or three-dimensional; images can be completely synthetic or can be produced by manipulating photographs.

Major Areas in Computer Graphics

- Major areas of Computer graphics :
  - Modelling :
    - Deals with the mathematical specification of shape and appearance properties in a way that can be stored on the computer.
    - For example, a coffee mug might be described as a set of ordered 3D points along with some interpolation rule to connect the points and a reflection model that describes how light interacts with the mug.
  - Rendering :
    - The term is inherited from art.
    - Deals with the creation of shaded images from 3D computer models.
  - Animation :
    - A technique to create an illusion of motion through sequences of images.
    - Describing how objects change in time .

Historical Milestones

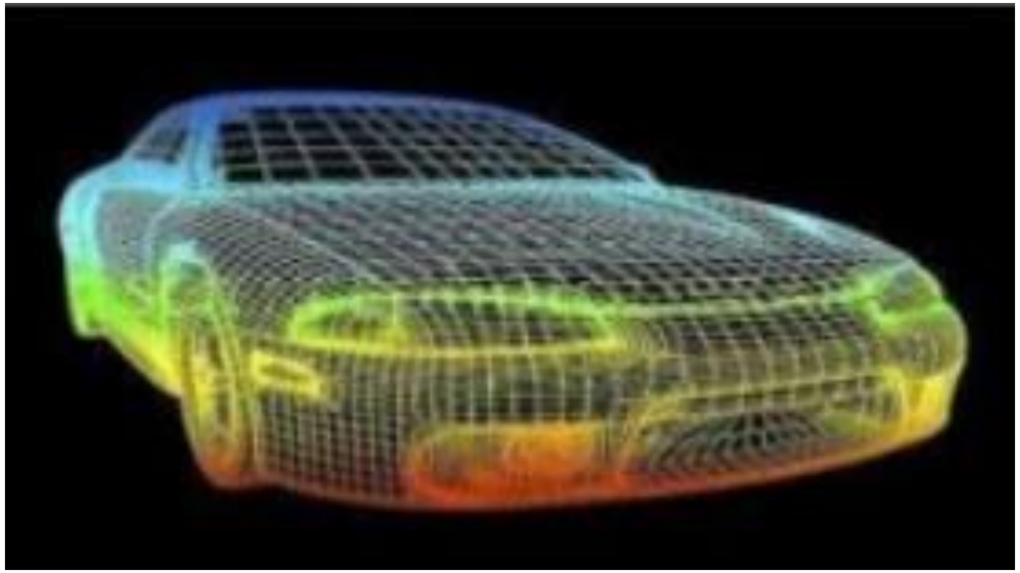
- 1950’s:
  - First Graphic image was created by using an oscilloscope to generate waveform artwork produced by manipulating the analog electronic beams.
- 1960’s:
  - Early theoretical development, mainly limited to research and military
  - 1962: Sketchpad (Ivan Sutherland)
- 1970’s:
  - ‘Traditional’ graphics pipeline developed
  - Driven by money from military simulation and automotive design industries
- 1980’s:
  - Many important core algorithms developed
  - Visual quality improved driven by demands from entertainment (movie) industry
  - 1985: Rendering Equation (James Kajiya)
- 1990’s:
  - Advanced algorithms developed as graphics theory matured
  - Broader focus on animation, data acquisition, NPR, physics…
  - 1995: Photon Mapping (Henrik Jensen)
- 2000’s:
  - Photoreal rendering evolves to the point of being able to render convincing images of arbitrarily
  - complex scenes on consumer hardware
  - Merging of computer graphics and computer vision
  - Cheap graphics hardware with vast capabilities, driven largely by video game industry

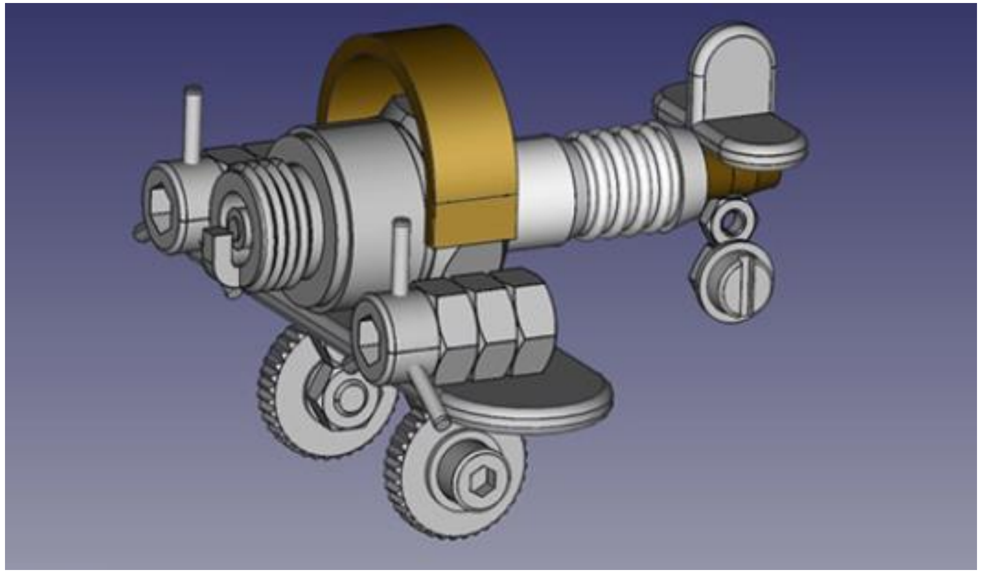
Application Areas in Computer Graphics

- Computer Aided Design(CAD)
- Presentation Graphics
- Education and Training
- Scientific Visualization
- Image Processing
- Medical Fields
- Entertainment
  - Movie
  - Gaming
- GUI

CAD/CAM

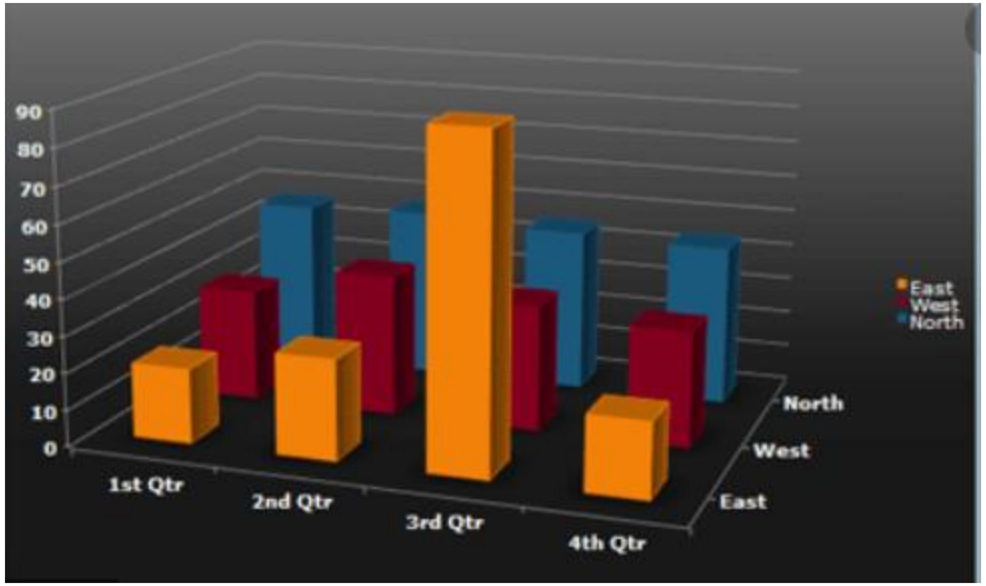
- Digitally create 2D drawings and 3D models of real-world products mostly for engineering and architectural systems





Presentation Graphics

- Used to summarize the mathematical, financial, scientific and economic data.
- Eg : Bar charts, Pie Charts, Line graphs



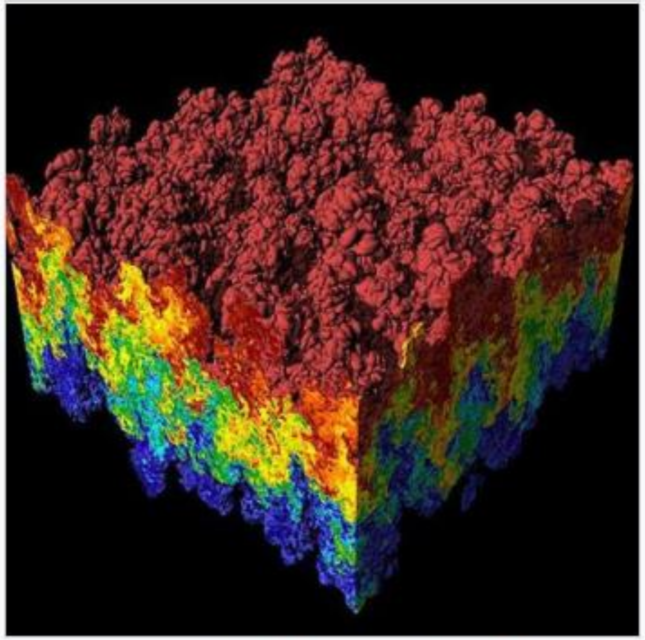
Education and Training

- Computer generated models of physical, financial and economic system are often used as educational aids.
- Various kinds of simulators program can be used to provide the trainings.
- E.g. automobile driving simulator.

Scientific Visualization

- Graphic techniques can be used to visualize large amount of data taken from different scientific domains.

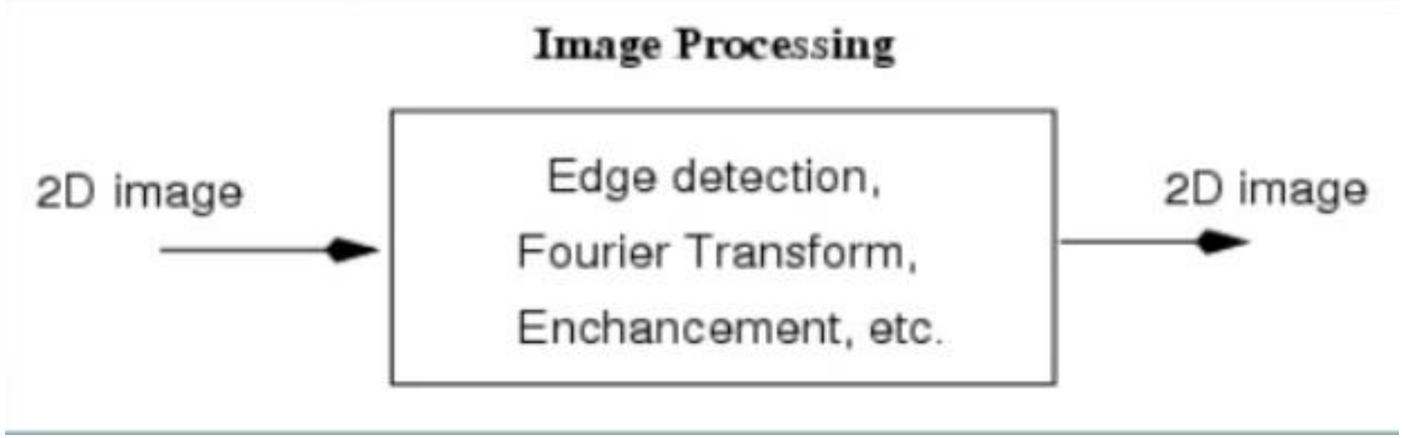
Scientific visualization of Fluid Flow: Surface waves in water



A scientific visualization of a simulation of a Rayleigh–Taylor instability caused by two mixing fluids.[1]

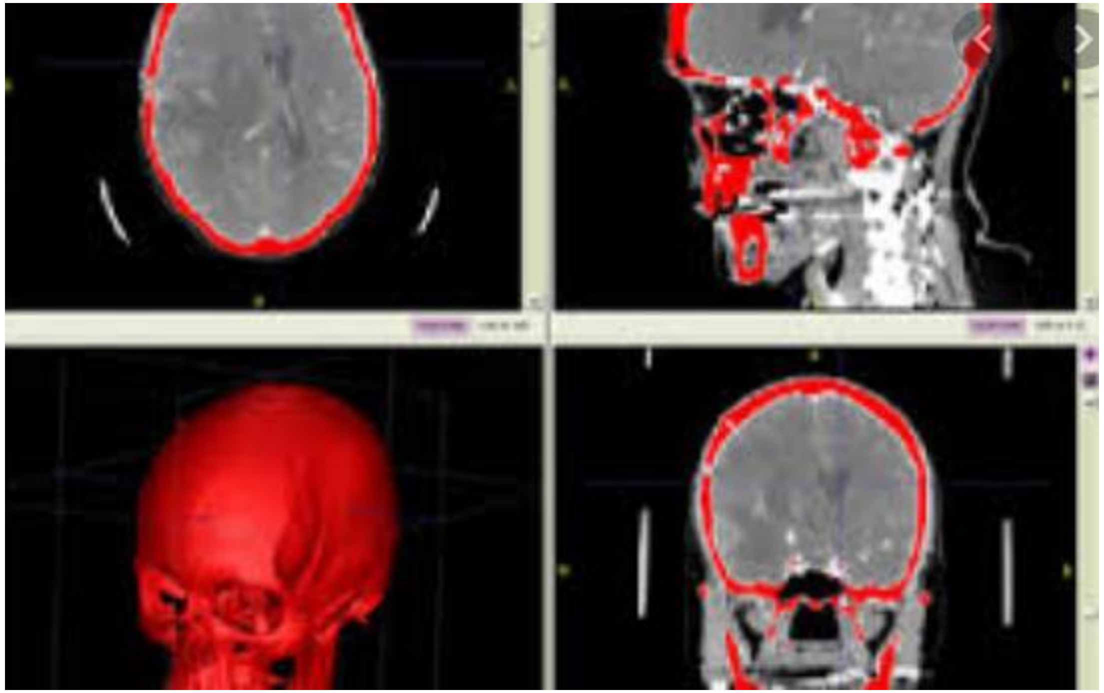
Image Processing

- Computer graphics is used to create pictures.
- Image processing applies techniques to modify or interpret the existing pictures.



Medical Fields

- Computer graphics can also be used to represent the various internal parts and process of the human body.

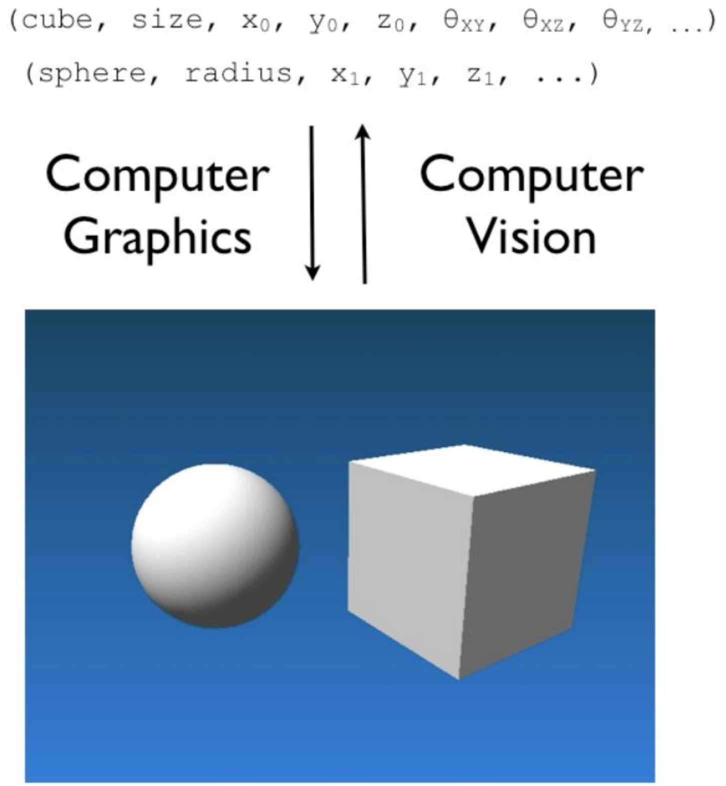


Entertainment

- Video Games
- Cartoons
- Animated Films

Computer Graphics Vs Computer vision

- An interdisciplinary field that deals with how computers can be made to gain high-level understanding from digital images or videos
- A central goal in computer vision is to take a set of 2D images (usually from a video or set of photos) and infer from that a 3D description of what is being viewed
- Vice versa of Computer Graphics



Raster Graphics

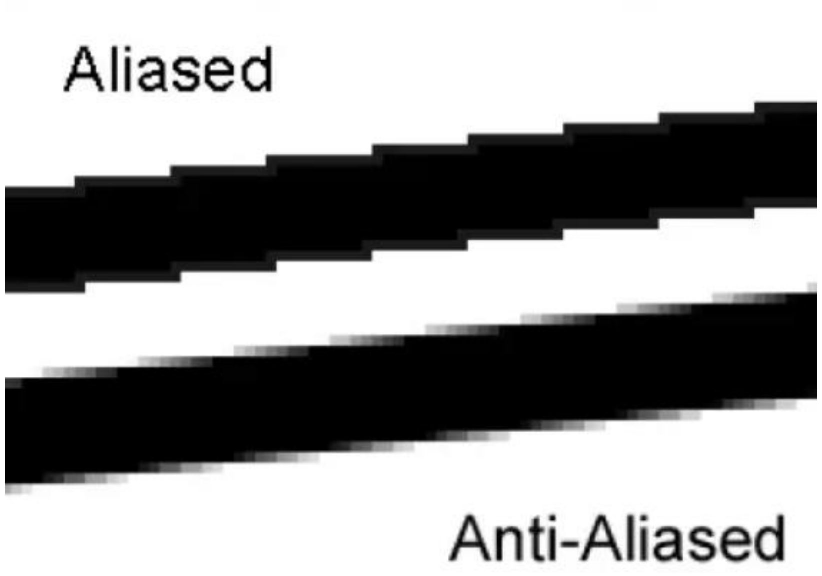
- Raster graphics are produced by using a grid of small squares know as ’pixels’. Each pixel is assigned a specific color value and a location.
- Often referred as ‘bitmap’ images.
- Quality is determined by the total number of pixels (Resolution) and the amount of information in each pixel (color depth).
- Framebuffer : Memory Area that stores the information(picture definitions and set of intensity values).
    - Black and white system : One bit per pixel is needed to control the intensity of screen positions.
    - True color/full color system : 24 bits per pixel is needed in frame buffer.

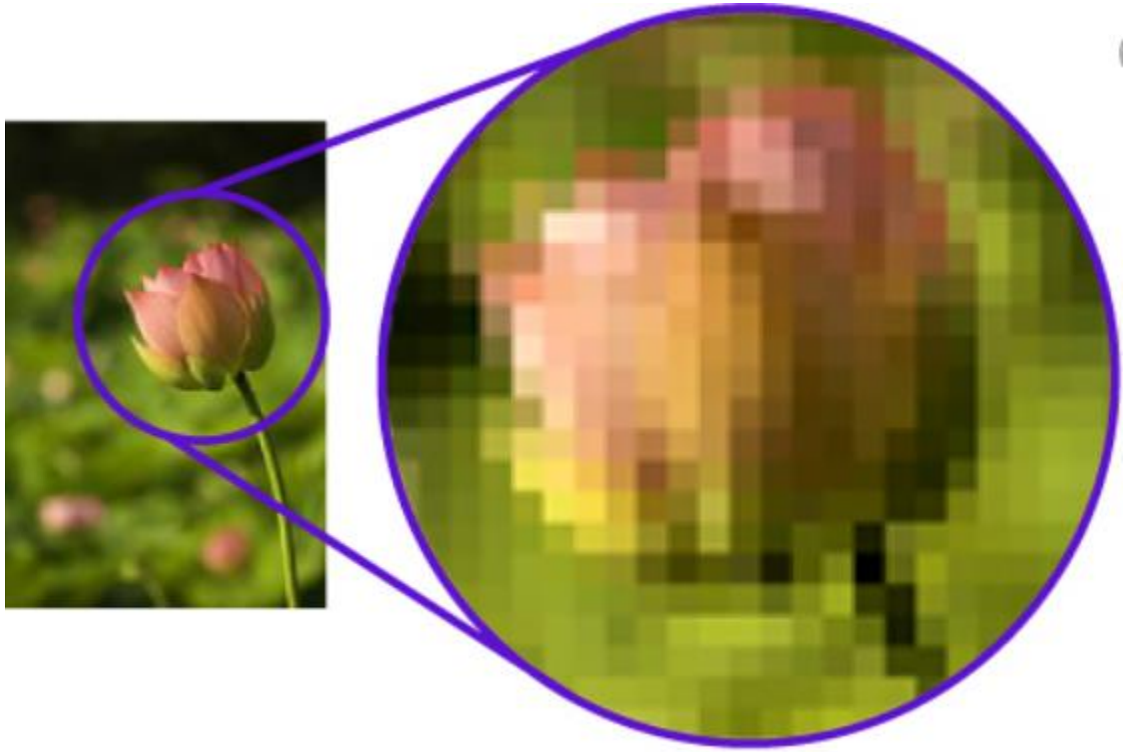
Raster Graphics Cont.

- Most Common file extensions of raster graphics are .jpg, .png, .gif, .bmp

Raster Graphics Cont.

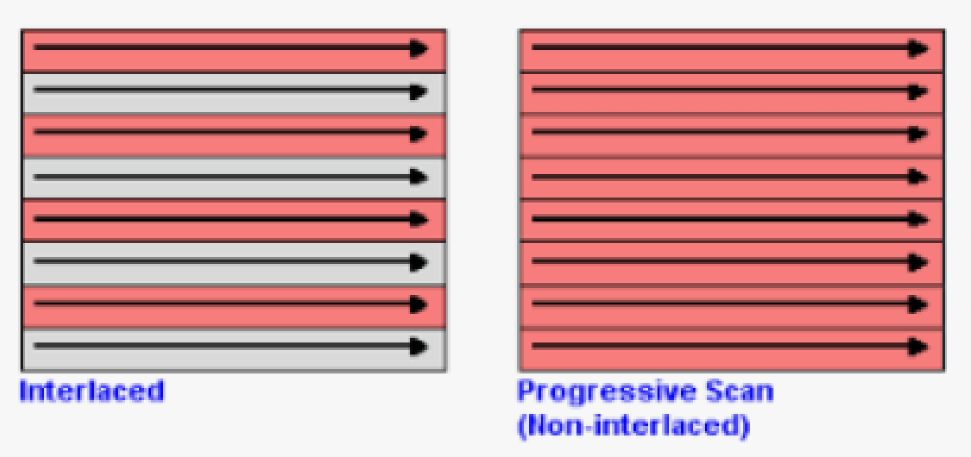
- When enlarges, it produces jagged lines that are plotted as discrete points.



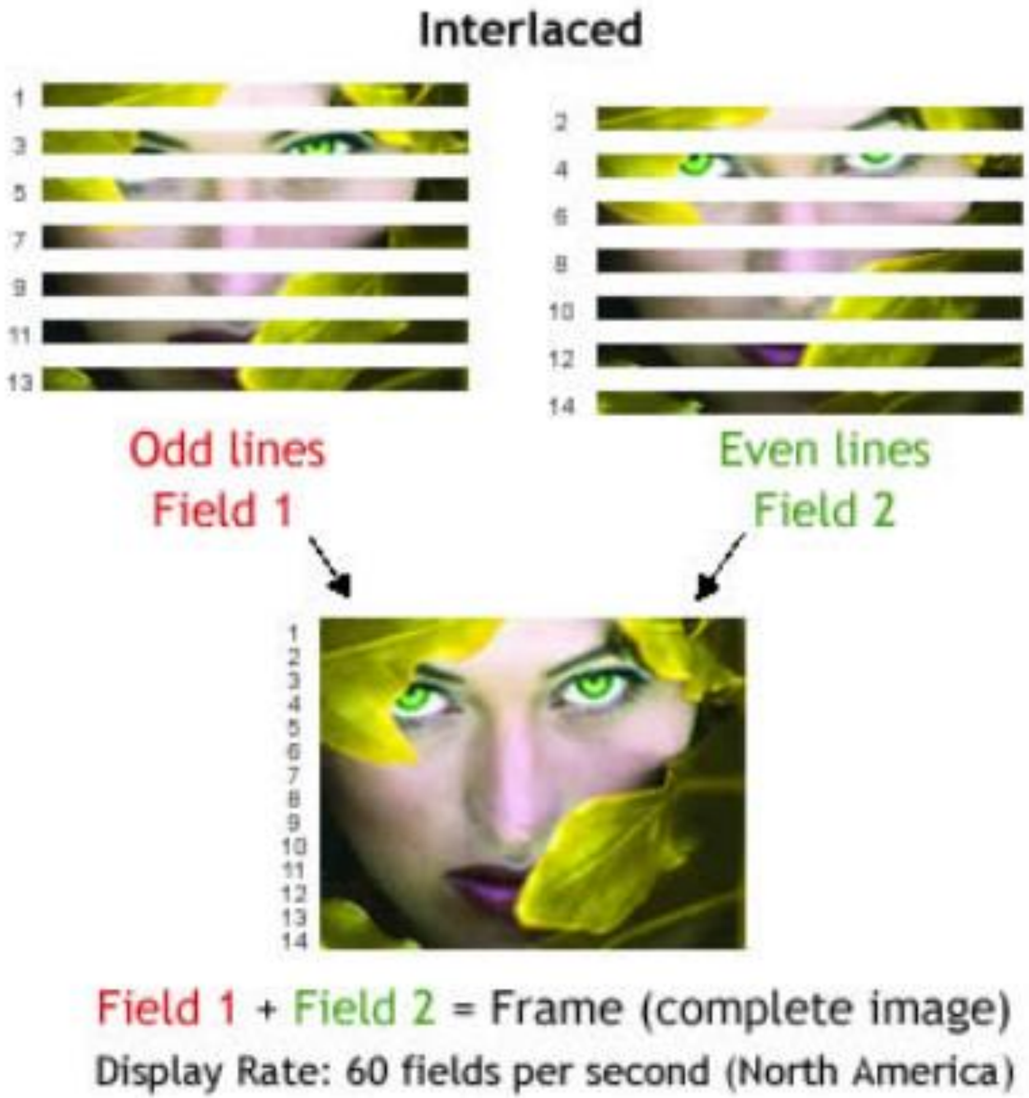


Interlacing

- Interlacing is a description of how the picture is created.
- With an interlaced display the picture is created by scanning every other line, and on the next scan, scanning every opposite line.
- Interlacing allows for a faster refresh rate by having less information during each scan at a lower cost.
- May cause flickering or noticeable line movements in some situations.

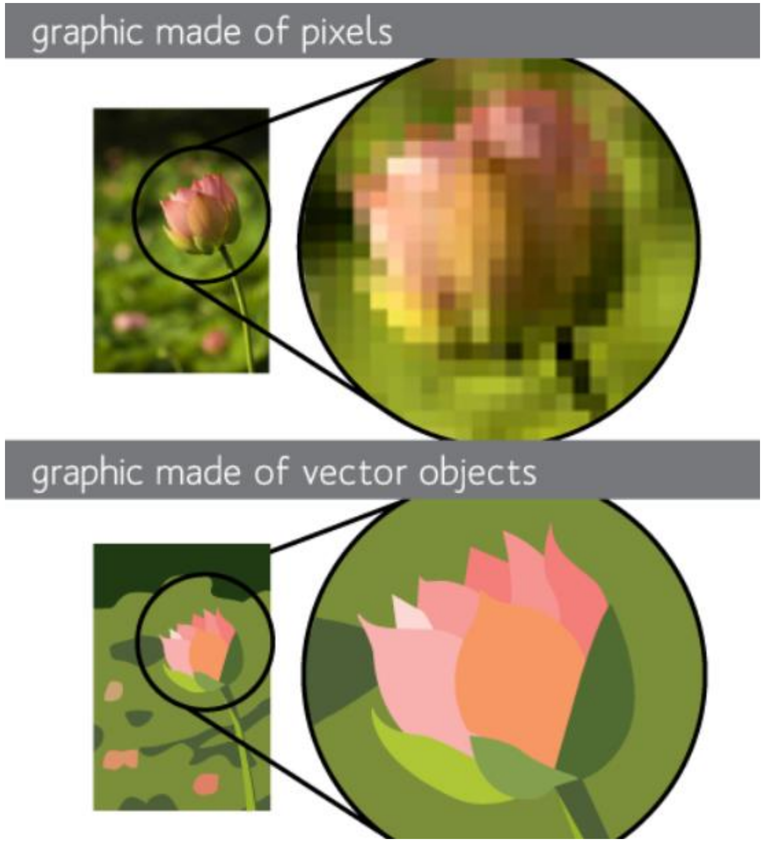


Interlacing Cont.



Vector Graphics

- Display by using geometrical primitives (lines, points, curves, polygons) which are based upon mathematical equations.
- Vector graphics can be moved or resized without losing quality or file size.
- Most Common file extensions of vector graphics are .svg, .ai, .cdr, .eps



Activity

- Compare and Contrast raster and vector graphics.

Graphics Pipeline

- Series of operations that takes 3D vertices/normals/triangles as input and generates fragments and pixels.
- Graphics pipeline refers to a series of interconnected stages through which data and commands related to a scene go through during rendering process.
- In traditional graphic pipeline following basic operations are performed;
  - Transformation
  - Lighting
  - Clipping
  - Scan conversion
  - Pixel processing

Activity

- Elaborate more in “Graphics Pipeline”.

Questions?